Amazon S3 – Practice Exercises

Exercise 1:

Create an Amazon S3 Bucket

In this exercise, you will create a new Amazon S3 bucket in your selected region. You will use this bucket in the following exercises.

1- Log in to the AWS Management Console.
2- Choose an appropriate region, such as S West (Oregon.)
3- Navigate to the Amazon S3 console. Notice that the region indicator now says Global. Remember that Amazon S3 buckets form a global namespace, even though each bucket is created in a specific region.
4- Start the create bucket process.
5- When prompted for Bucket Name, use **mynewbucket**
6- Choose a region, such as S West (Oregon.)
7- Try to create the bucket. You almost surely will get a message that the requested bucket name is not available. Remember that a bucket name must be unique globally.
8- Try again using your surname followed by a hyphen and then today's date in a six-digit format as the bucket name (a bucket name that is not likely to exist already.)

You should now have a new Amazon S3 bucket.

Exercise 2

Upload, Make Public, Rename and Delete Objects in your S3 Bucket

In this exercise, you will upload a new object to your S3 Bucket. You will then make this object public and view the object in your browser. You will then rename the object then finally delet it from the bucket.

Upload an Object

1- Load your new bucket in Amazon S3 console.
2- Select 'Upload' and then Add-Files.
3- Locate a file on your PC that you are okay with uploading to Amazon S3 and making public to the internet. (We suggest using a non-personal image file for the purpose of this exercise.)
4- Select a suitable file, then Start-upload. You will see the status of your file in Transfers section.

5- After your file is uploaded, the status should change to Done.

The file you uploaded is now stored as an Amazon S3 object and should be now listed in the contents of your bucket.

Open the Amazon S3 URL

6- Now open the properties for the object. The properties should include bucket, name and link.
7- Copy the Amazon S3 URL for the object.
8- Paste the URL in the address bar of a new browser window or tab.

You should get a message with an XML error code ***AccessDenied***. Even though the object has a URL, it is private by default, so it cannot be accessed by a web-browser.

Make the Object Public

9- Go back to the Amazon S3 console and select Make Public. (Equivalently, you can change the object's permissions and add grantee Everyone and permissions Open/Download.)
10- Copy the Amazon S3 URL again and try to open it n a browser or tab. Your public image-file should now display in the browser or browser tab.

Rename Object

11- In the Amazon S3 console, select Rename.
12- Rename the object, but keep the same file extension.
13- Copy the new Amazon S3 URL and try to open it in a browser or tab. You should see the same image file.

Delete the Object

14- In the Amazon S3 console, select Delete. Select OK when prompted if you want to delete the object.
15- The object has now been deleted.
16- To verify, try to reload the deleted object's Amazon S3 URL.

You should once again get the XML ***AccessDenied*** error message.

Exercise 3

Enable Version Control

In this exercise, you will enable version control on your newly created bucket.

Enable Versioning

1- In the Amazon S3 console, load the properties of your bucket. Don't open the bucket.
2- Enable versioning in the properties and select OK to verify. Your bucket now has versioning enabled. (Note that versioning can be suspended, but not turned off.)

Create Multiple Versions of an Object

3- Create a text file named foo.txt on your computer and write the word **blue** in the text file.
4- Save the text file to a location of your choosing.
5- Upload the text file to your bucket. This will be version 1.
6- After you have uploaded the text file to your bucket, open the copy on your local computer and change the word **blue** to **red**. Save the text file with the original filename.
7- Upload the modified file to your bucket.
8- Select show Versions on the uploaded object.

You will now see two different versions of the object with different Version IDs and possibly different sizes. Note that when you select Show Version, the Amazon S3 url now includes the version-ID in the query string after the object-name.

Exercise 4

Delete and Object and then Restore it.

In this exercise, you will delete an object in your Amazon S3 bucket and then restore it.

Delete an Object

1- Open the bucket containing the text file for which you now have two versions.
2- Select Hide Versions.
3- Select Delete, and then select OK to verify.
4- Your object will now be deleted and you can no longer see the object.
5- Select Show Versions.

Both versions of the object now show their version-IDs.

Restore an Object

6- Open your bucket.
7- Select Show Versions.
8- Select the oldest version and download the object. Note that the filename is simply foo.txt with no version indicator.
9- Upload foo.txt to the same bucket.
10- Select Hide Versions,and the file foo.txt should re-appear.

Exercise 5

Lifecycle Management

In this exercise you will explore the various options for lifecycle management.

1- Select your bucket in the AWS S3 console.
2- Under Properties, add Lifecycle Rule.
3- Explore the various options to add Lifecycle Rules to objects in this bucket. It is recommended that you do not implement any of these options, as you may incur additional costs. After you have finished, click the Cancel button.

Enable Static Hosting on Your Bucket

In this exercise, you will enable static hosting on your newly created bucket.

Note: Hosting a sample *index.html* file in our Amazon S3 Bucket.